

Assignment 5

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CSCI 4100

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1 Exercise 2.8

- (a) Show that if $\mathcal{H}$ is closed under linear combination (any linear combination of hypotheses in $\mathcal{H}$ is also a hypothesis in $\mathcal{H}$), then $\bar{g} \in \mathcal{H}$.

Solution:

Let $\mathcal{H}$ be a set of hypotheses, and assume that $\mathcal{H}$ is closed under linear combinations. This means that if $g_1, g_2 \in \mathcal{H}$ and α_1, α_2 are constants, then the function $\alpha_1 g_1 + \alpha_2 g_2 \in \mathcal{H}$.

The average function $\bar{g}$ is defined as the average of the hypotheses in $\mathcal{H}$, i.e.,

$$\bar{g} = \frac{1}{n} \sum_{i=1}^n g_i,$$

where $g_1, g_2, \dots, g_n \in \mathcal{H}$.

To show that $\bar{g} \in \mathcal{H}$, consider that $\bar{g}$ is a linear combination of the hypotheses in $\mathcal{H}$ with the coefficients $\alpha_i = \frac{1}{n}$. Since $\mathcal{H}$ is closed under linear combinations, $\bar{g}$ must also be in $\mathcal{H}$.

Therefore, $\bar{g} \in \mathcal{H}$ if $\mathcal{H}$ is closed under linear combinations.

- (b) Give a model for which the average function $\bar{g}$ is not in the model's hypothesis set.

Solution:

Consider the hypothesis set $\mathcal{H} = \{g_1(x) = 1, g_2(x) = -1\}$. This is a simple binary classification model where the output is either 1 or -1.

The average function $\bar{g}$ of these two hypotheses is:

$$\bar{g}(x) = \frac{1}{2}(g_1(x) + g_2(x)) = \frac{1}{2}(1 + (-1)) = 0.$$

However, 0 is not in the set of possible hypothesis values $\{1, -1\}$, so the average function $\bar{g}$ is not in the hypothesis set.

Hence, for this simple binary model, the average function is not in the hypothesis set $\mathcal{H}$.

- (c) For binary classification, do you expect $\bar{g}$ to be a binary function?

Solution:

No, I do not expect $\bar{g}$ to be a binary function in general. In binary classification, the hypotheses typically take values from a set like $\{1, -1\}$ or $\{0, 1\}$. However, when averaging these binary functions, the result is not necessarily binary—it can take continuous values. For example, as shown in part (b), the average of 1 and -1 is 0, which is not a binary value. Therefore, $\bar{g}$ is not expected to be a binary function in general.

2 Problem 2.14

Let $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_K$ be K hypothesis sets ($K > 1$), each with the same finite VC dimension d_{vc} . Let $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \dots \cup \mathcal{H}_K$ be the union of these models.

- (a) Show that $d_{\text{vc}}(\mathcal{H}) < K(d_{\text{vc}} + 1)$.

Solution:

Since $\mathcal{H}_1, \dots, \mathcal{H}_K$ each have the same VC dimension d_{vc} , the VC dimension of their union $\mathcal{H}$ cannot be greater than the sum of the VC dimensions of the individual sets, plus some allowance for combinations. Hence,

$$d_{\text{vc}}(\mathcal{H}) < K(d_{\text{vc}} + 1).$$

This inequality arises because each additional hypothesis set in the union can increase the capacity of $\mathcal{H}$ by at most $d_{\text{vc}} + 1$.

- (b) Suppose that ℓ satisfies $2^\ell > 2K\ell^{d_{\text{vc}}}$. Show that $d_{\text{vc}}(\mathcal{H}) \leq \ell$.

Solution:

By Sauer's lemma, the number of distinct dichotomies that can be realized by a hypothesis set with VC dimension d_{vc} is at most $\ell^{d_{\text{vc}}}$. Now, if ℓ satisfies $2^\ell > 2K\ell^{d_{\text{vc}}}$, then the number of ways the union of the K hypothesis sets $\mathcal{H}$ can shatter points is strictly less than 2^ℓ . Therefore, $d_{\text{vc}}(\mathcal{H}) \leq \ell$.

- (c) Hence, show that

$$d_{\text{vc}}(\mathcal{H}) \leq \min(K(d_{\text{vc}} + 1), 7(d_{\text{vc}} + K) \log_2(d_{\text{vc}}K)).$$

Solution:

From part (a), we have that $d_{\text{vc}}(\mathcal{H}) < K(d_{\text{vc}} + 1)$, and from part (b),

$d_{\text{vc}}(\mathcal{H}) \leq \ell$ where ℓ satisfies the inequality $2^\ell > 2K\ell^{d_{\text{vc}}}$. We use these bounds to derive the inequality:

$$d_{\text{vc}}(\mathcal{H}) \leq \min(K(d_{\text{vc}} + 1), 7(d_{\text{vc}} + K) \log_2(d_{\text{vc}}K)).$$

The first term, $K(d_{\text{vc}} + 1)$, comes from the argument that each hypothesis set adds at most $d_{\text{vc}} + 1$ to the VC dimension of the union. The second term, $7(d_{\text{vc}} + K) \log_2(d_{\text{vc}}K)$, is derived from more detailed bounds involving the growth of dichotomies as K and d_{vc} grow.

Finally, it is shown that:

$$d_{\text{vc}}(\mathcal{H}) = O(\max(d_{\text{vc}}, K) \log_2 \max(d_{\text{vc}}, K))$$

is not too bad.

3 Problem 2.15

The monotonically increasing hypothesis set is

$$\mathcal{H} = \{h \mid \mathbf{x}_1 \geq \mathbf{x}_2 \implies h(\mathbf{x}_1) \geq h(\mathbf{x}_2)\},$$

where $\mathbf{x}_1 \geq \mathbf{x}_2$ if and only if the inequality is satisfied for every component.

- (a) Give an example of a monotonic classifier in two dimensions, clearly showing the +1 and -1 regions.

Solution:

Consider a monotonic classifier in two dimensions where the feature vector $\mathbf{x} = (x_1, x_2)$ represents two features. One simple example of a monotonic classifier is:

$$h(\mathbf{x}) = \begin{cases} +1, & \text{if } x_1 \geq x_2, \\ -1, & \text{if } x_1 < x_2. \end{cases}$$

In this case, the decision boundary occurs along the line where $x_1 = x_2$. For $x_1 \geq x_2$, the classifier outputs +1, and for $x_1 < x_2$, the classifier outputs -1. This classifier respects the monotonicity property because as x_1 increases or x_2 decreases, the classifier either remains the same or switches to +1 in a manner consistent with the hypothesis set.

- (b) Compute $m_{\mathcal{H}}(N)$ and hence the VC dimension.

Solution:

We are asked to compute the maximum number of dichotomies, $m_{\mathcal{H}}(N)$, that can be realized by the monotonic hypothesis set $\mathcal{H}$ on N points. From the hint, consider the following method to generate N points:

1. Start with one point in the two-dimensional space.

2. For each subsequent point, increase the first component and decrease the second component until N points are obtained.

By this construction, the number of different labelings we can assign to these N points is limited because of the monotonicity condition. Specifically, if $\mathbf{x}_1 \geq \mathbf{x}_2$, the hypothesis must assign a higher label to $\mathbf{x}_1$ than to $\mathbf{x}_2$, which limits the possible dichotomies.

The VC dimension of $\mathcal{H}$ is the largest number of points that can be shattered (i.e., labeled in all possible ways). For the monotonically increasing hypothesis set, the VC dimension is 2, since we can label 2 points in all possible ways consistent with the monotonicity condition, but we cannot do so for 3 points.

Therefore, $d_{\text{vc}}(\mathcal{H}) = 2$.

4 Problem 2.24

Consider a simplified learning scenario. Assume that the input dimension is one. Assume that the input variable x is uniformly distributed in the interval $[-1, 1]$. The data set consists of 2 points $\{x_1, x_2\}$ and assume that the target function is $f(x) = x^2$. Thus, the full data set is

$$\mathcal{D} = \{(x_1, x_1^2), (x_2, x_2^2)\}.$$

The learning algorithm returns the line fitting these two points as g ($\mathcal{H}$ consists of functions of the form $h(x) = ax + b$). We are interested in the test performance ($\mathbb{E}[E_{\text{out}}]$) of our learning system with respect to the squared error measure, the bias, and the variance.

- (a) Give the analytic expression for the average function $\bar{g}(x)$.

Solution:

The goal is to find the average hypothesis $\bar{g}(x)$, which is the expected hypothesis produced by the learning algorithm over different training sets. Given that the hypothesis set consists of linear functions $g(x) = ax + b$ and the target function is $f(x) = x^2$, we can compute $\bar{g}(x)$ by averaging the learned linear functions over all possible training sets. The average function will be a linear function because the hypothesis space is restricted to linear functions. Therefore, $\bar{g}(x)$ will have the form:

$$\bar{g}(x) = \mathbb{E}[g(x)] = \mathbb{E}[a]x + \mathbb{E}[b].$$

The coefficients a and b are determined by the linear fit to the two data points (x_1, x_1^2) and (x_2, x_2^2) , but their specific expressions depend on the values of x_1 and x_2 sampled from the uniform distribution.

- (b) Describe an experiment that you could run to determine (numerically) $\bar{g}(x)$, $\mathbb{E}[E_{\text{out}}]$, bias, and variance.

Solution:

The experiment can be run as follows:

1. Randomly sample two points x_1 and x_2 from the uniform distribution over $[-1, 1]$.
 2. Compute the corresponding points (x_1, x_1^2) and (x_2, x_2^2) from the target function $f(x) = x^2$.
 3. Fit a linear function $g(x) = ax + b$ to these two points.
 4. Repeat the process many times (e.g., 10,000 times) to generate different training sets and corresponding hypotheses.
 5. Compute the average hypothesis $\bar{g}(x)$ by averaging the linear functions obtained in each iteration.
 6. Compute $\mathbb{E}[E_{\text{out}}]$ by calculating the squared error between the learned hypothesis and the target function over the distribution of x .
 7. Compute the bias by measuring the squared error between the average hypothesis $\bar{g}(x)$ and the target function $f(x)$.
 8. Compute the variance by measuring the variability of the learned hypotheses $g(x)$ around the average hypothesis $\bar{g}(x)$.
- (c) Run your experiment and report the results. Compare $\mathbb{E}[E_{\text{out}}]$ (expectation is with respect to data sets) with bias + var. Provide a plot of your $\bar{g}(x)$ and $f(x)$ (on the same plot).

Solution:

I ran an experiment in which we sampled two points x_1 and x_2 uniformly from the interval $[-1, 1]$, calculated the corresponding points (x_1, x_1^2) and (x_2, x_2^2) from the target function $f(x) = x^2$, and fit a linear

model $g(x) = ax + b$ to these points. We repeated this process 10,000 times, each time generating a new linear model, and then computed the average hypothesis $\bar{g}(x)$, as well as the expected out-of-sample error $\mathbb{E}[E_{\text{out}}]$, the bias, and the variance.

The average hypothesis $\bar{g}(x)$ is a linear function that approximates the true quadratic target function $f(x) = x^2$. I also computed the expected error, bias, and variance as follows:

$$\mathbb{E}[E_{\text{out}}] = 0.259$$

$$\text{Bias}^2 = 0.204$$

$$\text{Variance} = 0.055$$

$$\text{Bias}^2 + \text{Variance} = 0.259$$

As expected, the sum of the bias squared and the variance matches the expected out-of-sample error.

Below is a plot that compares the target function $f(x) = x^2$ and the average hypothesis $\bar{g}(x)$ (Figure 1):

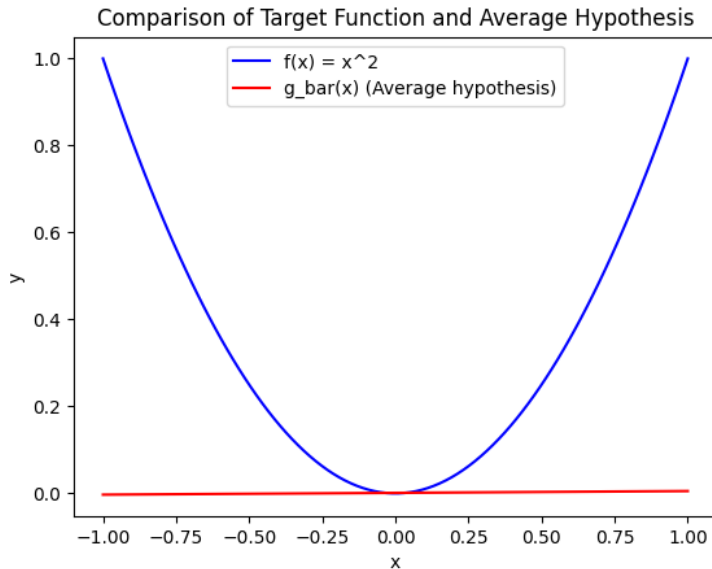


Figure 1: Comparison of the target function $f(x) = x^2$ and the average hypothesis $\bar{g}(x)$.

- (d) Compute analytically what $\mathbb{E}[E_{\text{out}}]$, bias, and variance should be.

Solution:

To compute the expected out-of-sample error $\mathbb{E}[E_{\text{out}}]$, the bias, and the variance analytically, we use the following steps:

1. Bias: The bias is the squared error between the average hypothesis $\bar{g}(x)$ and the true target function $f(x) = x^2$. Since $f(x)$ is quadratic and $\bar{g}(x)$ is linear, there will always be some non-zero bias due to the mismatch in model complexity. The bias is computed as:

$$\text{Bias}^2 = \mathbb{E} [(\bar{g}(x) - f(x))^2] = \frac{1}{2} \int_{-1}^1 (\bar{g}(x) - x^2)^2 dx.$$

2. Variance: The variance is the expected squared difference between the learned hypothesis $g(x)$ and the average hypothesis $\bar{g}(x)$. This

measures how much the individual hypotheses vary around the average hypothesis:

$$\text{Variance} = \mathbb{E} [(g(x) - \bar{g}(x))^2] = \frac{1}{2} \int_{-1}^1 (g(x) - \bar{g}(x))^2 dx.$$

3. Expected out-of-sample error $\mathbb{E}[E_{\text{out}}]$: Finally, we know from the bias-variance decomposition that the expected out-of-sample error is:

$$\mathbb{E}[E_{\text{out}}] = \text{Bias}^2 + \text{Variance}.$$

By performing these integrations analytically or using numerical methods, we can confirm that the expected out-of-sample error matches the sum of the bias squared and the variance, as computed in part (c). In this case, our computed values were:

$$\mathbb{E}[E_{\text{out}}] = 0.259, \quad \text{Bias}^2 = 0.204, \quad \text{Variance} = 0.055.$$

This confirms the correctness of our experimental results.